AUTOMATED SURVEY ANALYSIS TOOL

Minor Project Report

Project overview

The Automated Survey Analysis Tool is a Python-based application that processes customer survey responses from a CSV file. It calculates rating-based metrics, analyses written feedback using predefined keywords, identifies recurring complaint categories, and displays the results through a Streamlit dashboard.

Objectives

1. Read and validate customer survey data from CSV files.
2. Calculate average rating, rating distribution, and Customer Satisfaction Score (CSAT).
3. Categorise feedback into Positive, Negative, Mixed, and Neutral labels using keyword rules.
4. Identify common complaint areas and provide simple business recommendations.
5. Present results in an interactive dashboard and downloadable reports.

Tools and technologies

Python, CSV module, JSON, Streamlit, Matplotlib, and Pandas (for dashboard CSV export).

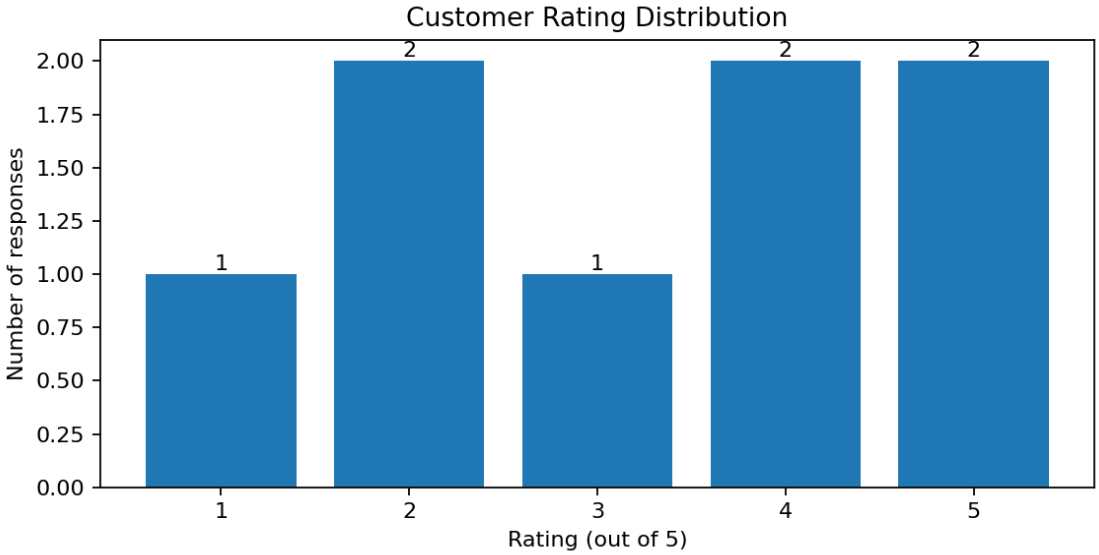
System workflow

CSV input → data validation → rating metrics → keyword-based feedback analysis → complaint category counts → JSON results → Streamlit dashboard.

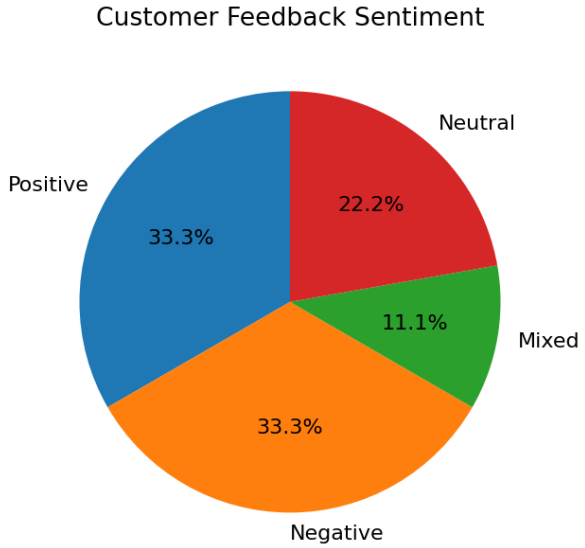
Sample dataset and results

Metric	Sample result
Total responses	9
Valid ratings	8
Missing ratings	1
Average rating	3.25/5
Five-star ratings	2
Low ratings (1 or 2)	3
CSAT (ratings 4 or 5)	50.0%

Rating distribution



Feedback sentiment



Complaint analysis and recommendations

• Staff behaviour: 1 comment(s)
• Delays and waiting: 2 comment(s)
• Customer support: 1 comment(s)
• Poor experience: 1 comment(s)

The most reported complaint category in this sample is Delays and waiting. The tool recommends reviewing delivery times and waiting processes. Overall sample CSAT is 50%, so the sample insight is that customer satisfaction needs improvement.

Methodology and limitations

The project uses predefined keyword lists and rule-based matching. Sentiment classification is not a trained machine-learning model and may misclassify context, negation, sarcasm, or words used in an unexpected way. Results depend on the keywords and categories configured in main.py. The sample dataset is small (9 responses), so its percentages should not be treated as representative of a wider customer population.

How to run

Install dependencies with `pip install -r requirements.txt`, run `python main.py` to generate the JSON analysis, and run `streamlit run dashboard.py` to open the dashboard. See README.md for Windows setup instructions.

Conclusion

The project demonstrates an end-to-end survey analysis workflow: reading structured data, validating inputs, calculating customer satisfaction metrics, analysing text feedback with keyword rules, and presenting business-oriented summaries in a dashboard. It can be extended with larger datasets, improved language processing, and public deployment.

PROJECT DEPLOYMENT & SUBMISSION UPDATE

Automated Survey Analysis Tool

This update supplements the project report with the final deployment details. The application has been successfully deployed and is accessible through the public Streamlit application link below.

Live application

https://automated-survey-analysis-tool-hkjj8x33ishyehu9zjcx6w.streamlit.app/

Source code repository

https://github.com/sathwickah007-lgtm/Automated-Survey-Analysis-Tool

Deployment configuration

The dashboard is deployed with Streamlit Community Cloud. The project dependency file is requirements.txt and includes Streamlit, Matplotlib, and Pandas. The live application link is also recorded in LIVE_DEMO_LINK.txt in the submission folder.

Final sample result note

The figures reported in this project—9 total responses, 8 valid ratings, 1 missing rating, average rating 3.25/5, and CSAT 50%—refer only to the included sample CSV. Sentiment labels are generated using predefined keyword rules; they are not predictions from a trained machine-learning model.

Future improvements

Possible future improvements include evaluating the keyword rules on a larger labelled dataset, improving handling of negation and context, adding more visual analytics, and expanding input validation. Public deployment is completed and is not a pending project task.